The Capability Trap: Why Self-Extracted Constraints Cannot Improve LLM Majority Voting

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Abstract

Recent work augments Large Language Model (LLM) reasoning with symbolic methods that extract first-order logic (FOL) constraints from reasoning traces and aggregate them via solvers such as MAX-SAT. We ask a practical question: when does such symbolic aggregation improve over simple majority voting (Self-Consistency, SC)? We instantiate the paradigm as SICA (Symbolic Invariant Consensus Aggregation), which extracts FOL constraints from sampled reasoning traces and selects answers via weighted MAX-SAT, and evaluate it across three reasoning domains (formal logic, commonsense, science) and five model families spanning instruction-tuned, distilled-reasoning, and full-reasoning architectures. Across all 18 tested conditions (combined N>1,300), SICA provides no significant accuracy gain over SC, with per-condition Δ ranging from -3.3pp to $+2.0$pp. We identify the structural cause: a *capability trap* in which self-extracted constraints achieve precision ≈ 0.50 (no better than chance) at distinguishing correct from incorrect answers, causing MAX-SAT to degenerate to weighted counting. A constraint ablation confirms that MAX-SAT, count-only, and random-weight aggregation produce statistically identical results (>99.6% agreement). We propose C_0 (multi-candidate rate) as a zero-cost diagnostic: C_0<5% predicts method degeneracy, while high C_0 is necessary but insufficient for gains. These findings provide practitioners with a principled early-stopping criterion before investing in symbolic aggregation overhead.

1 Introduction

Self-Consistency (SC) [Wang et al., 2023] improves LLM reasoning by sampling multiple reasoning traces and selecting the majority answer. A natural extension is to extract richer structure from these traces: formal constraints, logical invariants, or symbolic representations that a solver can aggregate more intelligently than simple voting [Pan et al., 2023, Olausson et al., 2023]. Several recent systems pursue this direction, using FOL translation, constraint satisfaction, or symbolic verification to rerank or select answers [Pan et al., 2023, Olausson et al., 2023, DyLoS, 2026, Aristotle, 2025, FoVer, 2025]. The appeal is clear: if reasoning traces contain more information than just the final answer, a solver that exploits that information should outperform voting.

But when does this actually work? Prior systems report gains on specific benchmarks, yet use trained verifiers, external oracles, or multi-paradigm solver selection. A fundamental question remains open: can the *same* LLM that generates reasoning traces also extract constraints of sufficient quality to improve its own voting? We call this the *self-extraction* setting, where a single model both reasons and formalizes.

We instantiate self-extraction as SICA (Symbolic Invariant Consensus Aggregation): sample K reasoning traces, extract FOL constraints from each via the same LLM, and solve a weighted MAX-SAT instance to select the best answer. We evaluate SICA across three reasoning domains (FOLIO, StrategyQA, ARC-Challenge, with GSM8K and ProofWriter as supplementary) and five model families (Qwen-14B, LLaMA-8B, Mistral-7B, DeepSeek-R1-Distill-7B, QwQ-32B), spanning instruction-tuned, distilled-reasoning, and full-reasoning architectures.

Our central finding is negative: SICA provides no significant improvement over SC in any tested condition. We identify the structural cause as a *capability trap*:

- SICA requires output diversity (multiple candidate answers) to differ from voting. We quantify this via C_0, the fraction of problems with more than one candidate answer across K traces.
- When C_0 is low (<5%, as with full reasoning models), SICA structurally degenerates to SC because there is only one candidate.
- When C_0 is high (38–62%, as with instruction-tuned models), self-extracted constraints achieve precision ≈ 0.50 at discriminating correct from incorrect answers, making MAX-SAT equivalent to counting.
- No tested condition achieves both sufficient diversity AND discriminative constraints simultaneously.

Our contributions are:

1. **Systematic negative result**: We show that self-extracted FOL constraint aggregation via MAX-SAT provides no significant accuracy gain over majority voting across 3 domains, 5 model families, and 18 experimental conditions (combined N>1,300).
2. **Capability trap characterization**: We identify and quantify the structural failure mode: self-extracted constraints achieve chance-level discriminative precision ($\bar{p} = 0.50$, $n = 307$ multi-candidate problems), causing MAX-SAT to degenerate to counting (confirmed by three-way ablation: MAX-SAT $\equiv$ count-only $\equiv$ random-weight, >99.6% agreement).
3. C_0 **deployment diagnostic**: We propose the multi-candidate rate C_0 as a zero-cost early-stopping criterion. C_0<5% reliably predicts degeneracy; high C_0 is necessary but insufficient for gains (Pearson $r = 0.05$, $p = 0.84$ across 18 conditions).

2 Related Work

Self-Consistency and extensions. SC [Wang et al., 2023] established majority voting over sampled reasoning traces as a strong baseline for LLM reasoning. Universal SC (USC) [Chen et al., 2023] extends this to free-form answers. Loo et al. [2025] demonstrate that SC scaling plateaus with increased samples, motivating richer aggregation. Confidence-weighted SC [Taubenfeld et al., 2025] assigns trace-level weights but remains within the voting paradigm. DiVeRSe [Li et al., 2023] combines diverse prompting with verification.

Neuro-symbolic reasoning. Logic-LM [Pan et al., 2023] and LINC [Olausson et al., 2023] translate natural language to FOL for symbolic solving, documenting that LLM-to-FOL translation is error-prone. Recent systems achieve gains using trained verifiers or multi-paradigm approaches: DyLoS [DyLoS, 2026] uses dynamic logic-solver selection, Aristotle [Aristotle, 2025] employs logic-complete decomposition, and FoVer [FoVer, 2025] trains a dedicated verifier. Our work differs by studying the specific failure mode of *same-model self-extraction* pipelines rather than these richer architectures.

Verification and reranking. Outcome Reward Models (ORM) [Cobbe et al., 2021] and Process Reward Models (PRM) [Lightman et al., 2023] use trained verifiers to rerank solutions. Math-Shepherd [Wang et al., 2024] extends process supervision. These approaches train separate verifiers, sidestepping the self-extraction bottleneck we characterize.

Test-time compute scaling. Snell et al. [2025] analyze optimal test-time compute allocation. DeepSeek-R1 [DeepSeek-AI, 2025] and OpenAI o1 [OpenAI, 2024] demonstrate that extended reasoning chains can substitute for external verification. Our finding that full reasoning models (QwQ-32B) produce near-zero diversity (C_0<5%) is consistent with this paradigm: models that "think longer" converge to consistent answers, eliminating the operating regime for symbolic aggregation.

Constraint satisfaction in NLP. SATBench [SATBench, 2025] evaluates LLM performance on satisfiability problems. Recent work explores constraint propagation with LLMs [LLM-CP, 2025]. Our MAX-SAT formulation draws on this line but applies it to answer aggregation rather than direct problem solving.

3 Method: SICA

SICA (Symbolic Invariant Consensus Aggregation) aggregates LLM reasoning traces via symbolic constraint satisfaction. The pipeline has four stages.

Stage 1: Trace generation. Given a problem q, sample K reasoning traces $\{t_1, \ldots, t_K\}$ from an LLM with temperature $\tau > 0$. Extract the final answer a_i from each trace t_i. Let $\mathcal{A} = \{a_1, \ldots, a_m\}$ be the set of distinct candidate answers.

Stage 2: Constraint extraction. For each trace t_i, prompt the same LLM to extract a set of FOL constraints $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,n_i}\}$ that formalize key reasoning steps. Each constraint $c_{i,j}$ is a FOL formula intended to capture an invariant of the reasoning process.

Stage 3: Constraint evaluation. For each candidate answer $a \in \mathcal{A}$ and each constraint $c_{i,j}$, evaluate whether a satisfies $c_{i,j}$. Define $\text{sat}(a, c_{i,j}) \in \{0, 1\}$. This produces a satisfaction matrix of size $|\mathcal{A}| \times |\mathcal{C}|$ where $\mathcal{C} = \bigcup_i \mathcal{C}_i$.

Stage 4: MAX-SAT aggregation. Formulate a weighted MAX-SAT instance where each constraint-answer pair contributes a weighted clause. The answer maximizing total satisfied constraint weight is selected:

$$a^* = \arg\max_{a \in \mathcal{A}} \sum_{c \in \mathcal{C}} w_c \cdot \text{sat}(a, c) \tag{1}$$

where w_c is the weight assigned to constraint c (default: uniform). In case of ties, SICA selects the answer supported by the trace with the highest individual constraint satisfaction count.

Baseline: Self-Consistency (SC). SC selects $a^* = \arg\max_{a \in \mathcal{A}} |\{i : a_i = a\}|$, using only final answers.

C_0 diagnostic. We define C_0 as the fraction of problems where $|\mathcal{A}| > 1$ (i.e., traces produce more than one distinct answer). When $C_0 = 0$, all traces agree and SICA trivially equals SC. C_0 is computable from Stage 1 alone, before any constraint extraction, making it a zero-cost diagnostic.

Table 1: Main results: SICA vs. SC across models and domains. Δ = SICA − SC in percentage points. C_0 = multi-candidate rate. All p-values from McNemar's test. No condition achieves significance at $\alpha = 0.05$.

Model	Domain	n	SICA Acc	SC Acc	Δpp	C_0%	p
Qwen2.5-14B	FOLIO+SQA+ARC	804	76.5	76.2	+0.25	38	–
Qwen2.5-14B	StrategyQA	60	56.7	53.3	+3.33	10	0.50
Qwen2.5-14B	ARC-Challenge	60	95.0	96.7	−1.67	5	1.00
Qwen2.5-14B	GSM8K	30	86.7	86.7	0.0	13	–
Qwen2.5-14B	GSM8K (expanded)	200	89.0	89.0	0.0	12	1.00
LLaMA-3.1-8B	StrategyQA	200	–	–	+2.0	62	0.424
LLaMA-3.1-8B	FOLIO	204	–	–	−2.45	62	0.267
DeepSeek-R1-7B	D012 (pilot)	60	68.3	71.7	−3.33	58	0.48
Qwen2.5-7B	D012 (pilot)	60	58.3	58.3	0.0	70	1.00
Qwen2.5-72B-AWQ	D012 (pilot)	60	80.0	80.0	0.0	22	1.00
QwQ-32B	All domains	46	–	–	0.0	2.6	–

4 Experiments

4.1 Setup

Domains. We evaluate on three primary domains: **FOLIO** (formal deductive logic, 204 problems), **StrategyQA** (commonsense reasoning, 200 problems for powered tests), and **ARC-Challenge** (science reasoning, 60 problems). Supplementary domains include **GSM8K** (arithmetic, 200 problems) and **ProofWriter** (included in QwQ-32B evaluation).

Models. We test five model families: (1) **Qwen2.5-14B-Instruct** as the primary model ($n = 804$ powered evaluation across domains); (2) **LLaMA-3.1-8B-Instruct** on StrategyQA ($n = 200$) and FOLIO ($n = 204$); (3) **DeepSeek-R1-Distill-Qwen-7B** ($n = 60$); (4) **QwQ-32B** as a full reasoning model ($n = 46$); (5) Within-family scaling with **Qwen2.5-7B** and **Qwen2.5-72B-AWQ**.

All experiments use $K = 12$ sampled traces unless otherwise noted.

Metrics. We report accuracy for both SICA and SC, their difference Δ (in percentage points), McNemar's test p-value for significance, and 95% bootstrap confidence intervals for powered tests. We also report C_0 (multi-candidate rate), constraint extraction rate, and constraint precision (fraction of constraints that correctly distinguish the gold answer from incorrect candidates).

4.2 Main Results

Table 1 summarizes results across all model-domain combinations. Across every tested condition, SICA provides no statistically significant improvement over SC.

Three patterns emerge from Table 1:

Instruction-scale models show high diversity but zero gain. Qwen-14B on its primary evaluation ($n = 804$) achieves $C_0 = 38\%$, meaning over a third of problems produce multiple candidate answers. Yet $\Delta = +0.25$pp, well within noise. LLaMA-8B shows the highest diversity ($C_0 = 62\%$) with powered tests ($n \geq 200$) yielding $\Delta = +2.0$pp on StrategyQA (CI: [−1.5, +5.5]) and $\Delta = -2.45$pp on FOLIO (CI: [−5.88, +0.98]), both non-significant and in opposite directions. The meta-analytic Δ across LLaMA-8B conditions is +0.3pp.

Full reasoning models produce near-zero diversity. QwQ-32B achieves an effective C_0 of 2.6% across all domains (AIME, ProofWriter, FOLIO), with $\Delta = 0.0$pp in every domain.

Table 2: Constraint ablation ($n = 804$, Qwen-14B). Three aggregation methods produce statistically identical results, confirming that MAX-SAT adds nothing when constraints lack discriminative power.

Aggregation	Accuracy	Agreement w/ MAX-SAT
MAX-SAT (full SICA)	76.49	–
Count-only	76.49	803/804
Random-weight	76.37	801/804
SC (majority vote)	76.24	–

Extended reasoning chains produce consistent answers, structurally eliminating the operating regime for SICA.

Within-family scaling confirms the trap from both ends. In the Qwen family, scaling from 7B ($C_0 = 70\%$, $\Delta = 0.0$) through 14B ($C_0 = 38\%$, $\Delta = +0.25$pp) to 72B ($C_0 = 22\%$, $\Delta = 0.0$) traces a capability-diversity tradeoff: weaker models produce more diversity but worse constraints, while stronger models produce better constraints but less diversity. Neither end benefits from SICA.

4.3 Constraint Ablation: MAX-SAT Adds Nothing Beyond Counting

If constraints carry discriminative information, the MAX-SAT solver should outperform simpler aggregation. We test this by comparing three aggregation strategies on the Qwen-14B $n = 804$ evaluation (Table 2).

MAX-SAT and count-only agree on 803 of 804 problems (McNemar $p = 1.0$). Random-weight agrees on 801/804 (McNemar $p = 0.82$). The three-way equivalence is the mechanistic signature of the capability trap: when constraint precision ≈ 0.50, each answer's satisfied constraint count is approximately proportional to its vote count. The MAX-SAT solver, despite its computational machinery, reduces to counting, which reduces to voting.

4.4 Per-Problem Constraint Analysis

To understand *why* constraints fail to discriminate, we analyze constraint precision at per-problem granularity across $n = 307$ multi-candidate problems from the Qwen-14B evaluation.

Constraint precision is indistinguishable from random. Mean constraint precision is 0.50 ($\sigma = 0.12$), with 89% of problems falling in the $[0.4, 0.6]$ range. There is zero correlation between problem-level diversity and constraint quality (Pearson $r = 0.03$, $p = 0.58$). Even among the 34 problems with precision > 0.6, SICA wins only 56% of the time (barely above chance).

Constraints between positive and negative flips. Problems where SICA flips from incorrect to correct (positive flips) have mean constraint precision 0.52; problems where SICA flips from correct to incorrect (negative flips) have precision 0.48. The difference is negligible, confirming that constraints do not systematically identify correct answers.

Qualitative analysis. We examine 8 representative problems in detail. Of extracted constraints, 75% are semantically meaningful FOL formulas, and 52% correctly evaluate against their source answer. However, only 12% are *discriminative*, producing different satisfaction values for different candidate answers. The typical failure mode is tautological constraints: a formula like $\forall x(P(x) \to Q(x))$ may be logically valid but evaluates to TRUE for all candidates (e.g., both "Proved" and "Unknown" in a deduction task). The extraction process produces logically correct but epistemically vacuous constraints.

Table 3: Weighted SC variants ($n = 804$, Qwen-14B). No weighting achieves significant improvement.

Weighting	Accuracy (%)	Δ vs. SC (pp)
Uniform (SC)	76.24	–
Inverse-length	76.74	+0.50
Trace-length	75.25	−0.99
Constraint-count	75.25	−0.99

4.5 C_0 as a Deployment Diagnostic

We validate C_0 as a predictor of SICA effectiveness across all 18 experimental conditions, spanning C_0 from 2.6% (QwQ-32B) to 70% (Qwen-7B).

C_0 is necessary but not sufficient. All conditions with $C_0 < 5\%$ show $\Delta = 0.0$pp exactly, confirming that low diversity makes SICA structurally degenerate. However, among conditions with $C_0 > 20\%$, the mean $|\Delta|$ is only 0.1pp with max $|\Delta| = 2.5$pp. The correlation between C_0 and Δ across all 18 conditions is $r = 0.05$ ($p = 0.84$), confirming that diversity provides no gradient of improvement.

Practical interpretation. C_0 functions as a gate, not a dial. Below 5%, practitioners can skip symbolic aggregation entirely with confidence. Above 5%, C_0 provides no guidance on whether gains will materialize, because the binding constraint shifts to extraction quality, which C_0 does not measure. Crucially, C_0 is computable from trace generation alone (Stage 1), before any constraint extraction, making it a zero-cost early-stopping criterion.

4.6 Tiebreak Analysis and K Ablation

Tiebreak is locally effective but globally irrelevant. On the 5 problems (out of 804) where SC produces an exact tie between candidates, SICA's constraint-based tiebreak achieves 80% accuracy vs. 50% for random tiebreak and 58% for alphabetical tiebreak. However, ties occur in only 0.6% of problems, contributing < 0.3pp to overall accuracy, which is statistically negligible.

K ablation. We test $K \in \{6, 8, 11, 12, 16, 20\}$ on Qwen-14B ($n = 60$). SICA $\leq$ SC at all values of K, including odd $K = 11$ where exact ties are structurally impossible. The $K = 11$ result ($\Delta = 0.0$pp) directly rules out the hypothesis that SICA's value lies solely in tiebreaking. Larger K marginally degrades SICA (e.g., $K = 20$: $\Delta = -1.7$pp), consistent with noise accumulation from additional non-discriminative constraints.

4.7 Additional Baselines

Weighted SC. We compare four weighting schemes on the $n = 804$ Qwen-14B data (Table 3). No scheme achieves significant improvement over uniform SC. Inverse-length weighting shows the largest (but non-significant) edge at +0.5pp ($p = 0.453$). This eliminates the concern that simpler weighting could succeed where MAX-SAT fails: the bottleneck is signal quality, not aggregation sophistication.

Logic-LM $\times$ K baseline. On FOLIO ($n = 204$), applying Logic-LM-style single-trace FOL verification K times and voting produces accuracy identical to both SC (79.4%) and SICA (79.4%). Whether using MAX-SAT (SICA), majority vote (SC), or single-trace verification

Table 4: Wall-time breakdown per problem ($n = 804$, Qwen-14B). SICA adds ∼49% overhead, dominated by LLM-based constraint extraction.

Stage	Mean time (s)	Fraction (%)
Trace generation	31.6	51.3
Constraint extraction	29.0	47.0
MAX-SAT solving	0.003	<0.01
Total	61.6	100.0

(Logic-LM $\times K$), same-model extraction produces equivalent results. The bottleneck is extraction quality, not aggregation mechanism.

4.8 Overhead Analysis

SICA nearly doubles inference cost for negligible accuracy gain (Table 4).

The overhead is dominated by LLM calls for constraint extraction (47% of total time), while MAX-SAT solving itself is negligible (<0.01%). The cost multiplier is 1.95× compared to SC alone. Combined with zero effectiveness, this definitively answers the practitioner question: SICA is not worth the overhead in any tested condition.

5 Discussion

Anatomy of the capability trap. The capability trap has a precise mechanistic explanation. Self-extracted constraints are logically correct in 75% of cases and correctly evaluate against their source answer 52% of the time. However, only 12% of constraints are *discriminative*: they produce different satisfaction outcomes for different candidate answers. The remaining 88% are tautological or overly general, evaluating identically across all candidates. When most constraints are non-discriminative, the MAX-SAT objective $\sum_c w_c \cdot \text{sat}(a, c)$ becomes proportional to the number of supporting traces, reducing the solver to a weighted vote. The three-way equivalence of MAX-SAT, count-only, and random-weight aggregation (Section 4.3) is the direct consequence: constraint weights carry no information when the constraints themselves do not discriminate.

Why does self-extraction fail? The extraction process asks the LLM to formalize its own reasoning steps as FOL constraints. The resulting formulas tend to capture general reasoning patterns (e.g., modus ponens applications, domain axioms) that are valid regardless of the specific answer. This is structurally different from constraints designed to test specific answer candidates, which would require the model to reason *about* its own uncertainty, a metacognitive capability that current LLMs lack in the self-extraction setting.

Within-family scaling. The Qwen scaling results (7B: $C_0 = 70\%$, 14B: $C_0 = 38\%$, 72B: $C_0 = 22\%$) reveal that capability and diversity trade off within a model family. Weaker models produce more diverse (and thus more frequently disagreeing) outputs but extract worse constraints (precision 0.49 for 7B vs. 0.51 for 72B). Stronger models extract marginally better constraints but produce insufficient diversity to exercise them. This tradeoff is structural: it arises from the same capability that determines both answer consistency and constraint quality.

Relationship to published systems. Systems like DyLoS, Aristotle, FoVer, and LogicAgent achieve gains on benchmarks overlapping with ours. The key distinction is architectural: these systems employ trained verifiers, external oracles, logic-complete decomposition, or multi-paradigm solver selection. Our finding characterizes the specific failure mode of *same-model*

self-extraction pipelines, not symbolic reasoning broadly. The capability trap may be breakable by decoupling the extraction and reasoning models, as these systems implicitly do.

Reasoning models bypass the need entirely. QwQ-32B, as a full reasoning model, achieves $C_0 = 2.6\%$, making SICA structurally degenerate. This is consistent with the test-time compute scaling paradigm [Snell et al., 2025, DeepSeek-AI, 2025]: models that invest compute in extended chains achieve consistency through deliberation rather than aggregation. For reasoning models, the question of symbolic aggregation is moot.

Practical recommendations. Practitioners considering symbolic aggregation over self-extracted constraints should: (1) Compute C_0 from a small pilot ($n \geq 30$, $K = 12$). If $C_0 < 5\%$, the method will degenerate to SC. (2) Even if $C_0 > 20\%$, do not expect gains without evidence that the extraction model produces discriminative constraints (precision $\gg 0.50$). (3) Consider the $\sim 2\times$ overhead: SC achieves the same accuracy at half the cost.

6 Conclusion

We presented a systematic evaluation of symbolic constraint aggregation via self-extracted FOL constraints (SICA) across 3 reasoning domains, 5 model families, and 18 experimental conditions. Our central finding is a *capability trap*: self-extracted constraints achieve chance-level discriminative precision, causing MAX-SAT to degenerate to counting and providing no improvement over simple majority voting. We validated this through constraint ablation (MAX-SAT $\equiv$ count-only $\equiv$ random-weight), per-problem precision analysis ($n = 307$, mean precision $= 0.50$), and cross-condition C0 analysis ($r = 0.05$ between diversity and gain). The multi-candidate rate C_0 provides a zero-cost diagnostic for practitioners: it reliably predicts when symbolic aggregation cannot help, potentially saving significant inference overhead ($\sim 2\times$ cost).

Limitations. Our study is scoped to same-model self-extraction pipelines. Systems using trained verifiers, external oracles, or cross-model extraction may not share this limitation. We test one MAX-SAT formulation; alternative symbolic aggregation strategies (e.g., SAT, SMT, answer set programming) could yield different results. GSM8K and ARC-Challenge evaluations use smaller samples ($n \leq 200$); powered replications on these domains would strengthen conclusions.

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A K Ablation Details

Table 5 shows SICA vs. SC across K values on Qwen-14B ($n = 60$, D012 subset).

Table 5: K ablation (Qwen-14B, $n = 60$). SICA $\leq$ SC at all K values.

K	SICA Acc (%)	SC Acc (%)	Δpp
6	73.3	73.3	0.0
8	73.3	73.3	0.0
11	73.3	73.3	0.0
12	73.3	75.0	−1.7
16	71.7	73.3	−1.7
20	70.0	71.7	−1.7

B Tiebreak Baseline Details

On the $n = 804$ Qwen-14B evaluation, only 5 problems (0.6%) produce exact SC ties. Table 6 compares tiebreak strategies on these tied problems.

Table 6: Tiebreak accuracy on 5 tied problems (Qwen-14B, $n = 804$).

Tiebreak Strategy	Accuracy (%)
SICA (constraint-based)	80.0
Alphabetical	58.0
Random	50.0
First-seen	42.0

SICA's constraint-based tiebreak is locally effective but contributes < 0.3pp to overall accuracy due to the rarity of ties.

C Overhead Breakdown

The $\sim 2\times$ cost multiplier ($1.95\times$) of SICA over SC is dominated by constraint extraction LLM calls (47.0% of total wall time). MAX-SAT solving accounts for $<0.01\%$ of computation. This cost structure means that any optimization of SICA's overhead must target the extraction step, not the solver.

D Within-Family Scaling

Table 7: Within-family scaling (Qwen, $n = 60$, $K = 12$). C_0 and constraint quality trade off, maintaining zero net gain.

Model	SC (%)	Sica (%)	Δpp	C_0 (%)	Extr.	Prec.
Qwen-7B	58.3	58.3	0.0	70	0.78	0.49
Qwen-14B	73.3	73.3	0.0	38	–	–
Qwen-72B	80.0	80.0	0.0	22	0.94	0.51

The scaling data reveals a structural tradeoff: weaker models produce higher diversity (C_0) but lower constraint quality (extraction rate and precision), while stronger models improve constraint quality at the expense of diversity. At no point along this spectrum does Sica outperform SC.

E Logic-LM Comparison

On FOLIO ($n = 204$), Logic-LM-style single-trace FOL verification applied $K = 12$ times with majority voting achieves 79.4% accuracy, identical to both SC (79.4%) and Sica (79.4%). Logic-LM single-trace accuracy is 78.9%. The equivalence across all three aggregation paradigms confirms that same-model extraction quality is the binding bottleneck.

F QwQ-32B Details

QwQ-32B produces near-deterministic outputs across all domains: effective $C_0 = 2.6\%$ (only 1 out of 46 problems produces multiple candidates). Per-domain results: AIME $\Delta = 0.0$pp, ProofWriter $\Delta = 0.0$pp, FOLIO $\Delta = 0.0$pp. Sica structurally degenerates to SC for full reasoning models.

G DeepSeek-R1-Distill-7B Details

Despite being distilled from a reasoning model, DeepSeek-R1-Distill-Qwen-7B ($n = 60$) behaves as an instruction-scale model: $C_0 = 58.3\%$, comparable to other 7–8B instruction models. The 7B capacity is insufficient to maintain deterministic reasoning chains. $\Delta = -3.33$pp (n.s., McNemar $p = 0.48$, 0 Sica wins vs. 2 SC wins among discordant pairs). 72 traces were truncated due to context length, potentially contributing to the negative direction.